Kumar Manthan

rishu.km04@gmail.com
9931710087
linkedin.com/in/kumar-manthan-05b29228b
github.com/Dracon-24
Chennai, India



EDUCATION

Vellore Institute of Technology, Chennai — **CGPA 8.37/10**
B.Tech Computer Science and Engineering (AI & ML) — 2023 - 2027

SKILLS

Programming: C++ (Primary), C, Java, JavaScript, Python

Frontend: HTML, CSS, Vanilla JavaScript

Backend: Node.js, Express.js, REST APIs

Database: PostgreSQL (Primary), MongoDB

Authentication & Security: JWT, bcrypt, OAuth 2.0, SHA-256, Ed25519

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

PROJECTS

Posturector
Personalized Adaptive Lumbar Posture Monitoring & Correctoin System
Tech Stack: React Native, Expo, TypeScript, Node.js, Express.js, SQLite, IMU Sensors, REST APIs

- Developed a **real-time** lumbar posture monitoring system using **IMU sensors**, accelerometer - gyroscope sensor fusion, and complementary filtering.
- Implemented **personalized calibration**, **baseline relative deviation detection**, stability & persistence filtering, and **real time haptic feedback** to reduce false corrections.
- Designed a **behavioral adaptive threshold controller** using five longitudinal posture features and multi timescale EMA based evidence accumulation.
- Integrated **Borg RPE** and **EMG based fatigue** measurements to learn data driven weights for the five behavioral features and improve personalized adaptive threshold decisions.
- Implemented bounded, reversible threshold adaptation with severe deviation & instability safety gates and **adaptive feedback** based on correction response history.

MediChain
Secure Medical Record Integrity System
Tech Stack: Python, Flask, SQLite, JavaScript, HTML, CSS, Cryptography, REST APIs

- Developed an **offline lightweight medical record integrity system** using **SHA 256 hashing** and **Ed25519 digital signatures** to verify record authenticity and detect unauthorized modifications.
- Implemented **nonce based challenge response authentication** and password encrypted private key storage for staff credentials.
- Designed **hash linked record versioning** and **tamper evident audit logs** to track updates and verify stored history.
- Built **REST APIs** and a web interface for staff registration, record creation, modification, version history, and integrity verification.

Touroholic
Full Stack Tour Discovery, Booking & Review Management Platform
Tech Stack: Node.js, Express.js, MongoDB, Mongoose, JavaScript, Leaflet.js, REST APIs, JWT

- Developed a full stack **tour discovery and booking platform** with responsive tour listings, detailed itineraries, user profiles, and personalized booking history and a complete review and rating system.
- Designed the application using **MVC architecture** and built **RESTful APIs** for managing tours, users, bookings, reviews, and authentication.
- Implemented **JWT based authentication** with account registration, password hashing and recovery, profile management, and role based access control.
- Integrated interactive **Leaflet and OpenStreetMap** maps to visualize tour locations and itinerary stops.

COMPETITIVE PROGRAMMING

Leetcode
Solved 500+ DSA problems | Knight (Max Rating: 1866)
Solved 50+ SQL problems

CodeChef
Solved 100+ DSA problems | 2 Star Coder (Max Rating: 1454)

GeeksforGeeks
Solved 150+ DSA problems